Late Refill Risk Prediction

Data-AI-Hackathon – Task 1, Challenge A

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Team Submission

March 31, 2026

Time-based split + patient holdout

1. Problem and grouping challenge

Goal

Predict **late refill** risk from historical pharmacy claims.

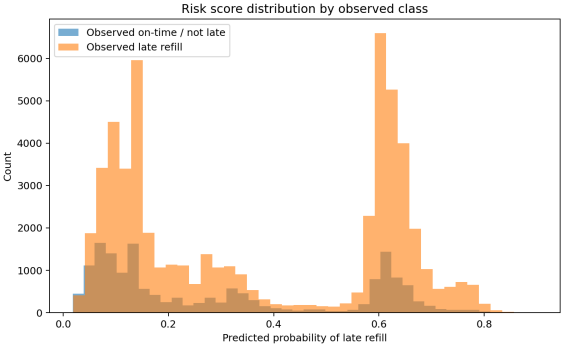
Main data challenge

- `PROD_SRVC_ID` is highly specific
- overly narrow grouping fragments refill chains
- tested **NDC11, NDC8, NDC7**

Chain definition

$$\hat{t}_{i+1} = t_i + d_i \qquad t_{i+1} > t_i + d_i + 7$$

- patient + grouped drug
- ordered by service date



2. Feature design

Main feature groups

- drug grouping
- current fill
- chain dynamics
- refill gap history
- patient utilization history
- payment and cost history
- same-day fill context
- calendar / time features

Examples

- `DAYS_SUPLY_NUM`, `QTY_DSPNSD_NUM`
- chain length, fill index, fraction elapsed
- prior gaps, mean prior gap

Dataset columns
132

Numeric
111

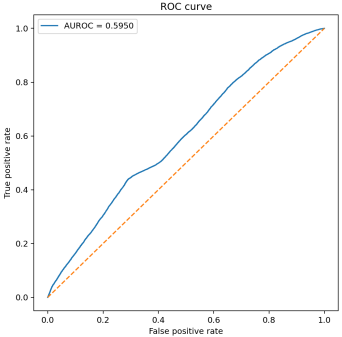
Categorical
4

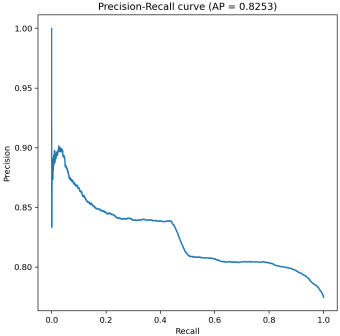
Model inputs
115

Categorical fields:
NDC11, NDC7, NDC8, PROD_SRVC_ID

3. Final XGBoost results

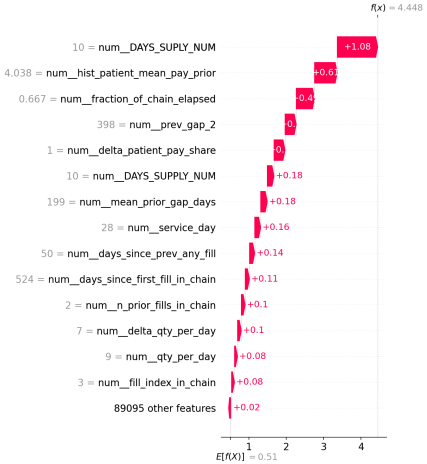
AUROC	PR-AUC	F1
0.6357	**0.8430**	**0.7496**





Validation setup: weighted training, best threshold = 0.44, validation prevalence = 0.7746.

4. Model comparison and interpretation



Models tested

- LightGBM
- XGBoost
- FT-Transformer

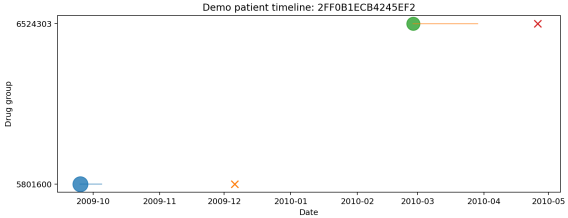
Best FT-Transformer

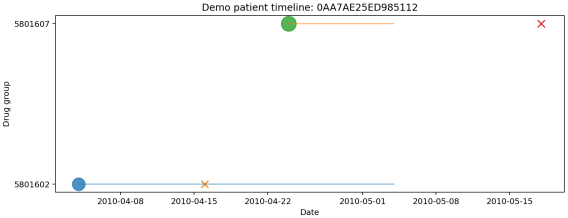
- AUROC: **0.6095**
- PR-AUC: **0.8313**
- F1: **0.8791**
- Brier: **0.1674**

Top drivers

- days supply
- prior patient pay

5. Patient demo





NDC7 gave the most stable refill chains and the clearest patient-level interpretation.